

新南方烹饪学校General Chemistry

2023春期中

This test is closed-book, closed-notes, and closed-calculators. A periodic table, table of fundamental constants and electrochemical series are provided. Note: please write down your answers on the answer sheet.

Section A: Multiple Choice - Select the one that is best in each case. (40 pts)

Number of questions: 20

Points per question: 2

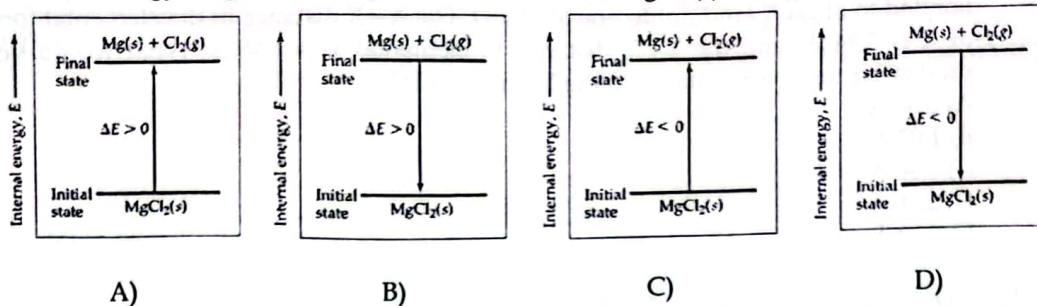
- Instructor: _____
- SID: _____
- Name: _____
- A1).** Blend Water, sugar, some amount of ethyl alcohol, and grape juice together uniformly and give clear and tasty wine. Which term could be used to describe this sample of wine?
- A) heterogeneous mixture
 - B) solution
 - C) pure chemical substance
 - D) compound
 - E) element
- A2).** Which of these changes involved a physical change and not a chemical change?
- A) burning of natural gas
 - B) formation of ethanol from glucose by yeast
 - C) formation of carbon dioxide gas from glucose
 - D) evaporation of ethanol
 - E) formation of carbon deposit during ethanol distillation
- A3).** Use appropriate metric prefixes to write the measurement $(1.45+1.2 \times 1.85) \times 10^{-12} \text{ m}$ without use of exponents.
- A) 367 nm
 - B) 3.67 pm
 - C) 3.67 μm
 - D) 3.7 pm
 - E) 367 μm
- A4).** In 1909, Ernest Rutherford performed an experiment to explore the atomic structure. How do the results of the Rutherford experiment suggest that the nucleus is positively charged?
- A) Because the center spot disappears when gold is placed in front of the alpha particle beam.
 - B) The Rutherford experiment can't tell us the nucleus is positively charged.
 - C) A small amount of scattering was observed at large angles, with some particles scattered back in the direction from which they had come.
 - D) Because some of the alpha particles are scattered at wide angles.
 - E) Most alpha particles undergo no scattering because most of the atom is empty.
- A5).** In which of the following species is the number of protons less than the number of electrons?
- A) Mn
 - B) Ce^{4+}
 - C) Ca^{2+}
 - D) S^{2-}
 - E) Ti^{2+}



- A6). Determine the most conventional balanced equation for the reaction that occurs when ethylene glycol, $\text{C}_2\text{H}_4(\text{OH})_2$, burns in air.
- A) $\text{C}_2\text{H}_4(\text{OH})_2(l) + 3 \text{O}_2(g) \rightarrow 2 \text{CO}_2(g) + 3 \text{H}_2\text{O}(g)$
 - B) $\text{C}_2\text{H}_4(\text{OH})_2(l) + 5/2 \text{O}_2(g) \rightarrow 2 \text{CO}_2(g) + 3 \text{H}_2\text{O}(g)$
 - C) $4 \text{C}_2\text{H}_4(\text{OH})_2(l) + 10 \text{O}_2(g) \rightarrow 8 \text{CO}_2(g) + 12 \text{H}_2\text{O}(g)$
 - D) $2 \text{C}_2\text{H}_4(\text{OH})_2(l) + 5 \text{O}_2(g) \rightarrow 4 \text{CO}_2(g) + 6 \text{H}_2\text{O}(g)$
 - E) $\text{C}_2\text{H}_4(\text{OH})_2(l) + 5 \text{O}(g) \rightarrow 2 \text{CO}_2(g) + 3 \text{H}_2\text{O}(g)$
- A7). The compound dioxane, which is used as a solvent in various industrial processes, is composed of C, H, and O atoms. Combustion of a 2.203-g sample of this compound produces 4.401 g CO_2 and 1.802 g H_2O . A separate experiment shows that it has a molar mass of 88.1 g/mol. Which of the following is the correct molecular formula for dioxane
- A) $\text{C}_2\text{H}_4\text{O}$
 - B) CH_2
 - C) $\text{C}_4\text{H}_8\text{O}_2$
 - D) $\text{C}_4\text{H}_4\text{O}_2$
 - E) CH_4O
- A8). Which compound has the atom with the highest oxidation number?
- A) CaS
 - B) Na_3N
 - C) MgSO_3
 - D) $\text{Al}(\text{NO}_2)_3$
 - E) NH_4Cl
- A9). Aqueous solutions of a compound did not form precipitates with Cl^- , Br^- , I^- , SO_4^{2-} , CO_3^{2-} , PO_4^{3-} , OH^- , or S^{2-} . This highly water-soluble compound produced the foul-smelling gas H_2S when the solution was acidified. This compound is _____.
- A) $\text{Pb}(\text{NO}_3)_2$
 - B) $(\text{NH}_4)_2\text{S}$
 - C) KBr
 - D) Li_2CO_3
 - E) AgNO_3
- A10). Based on the activity series, which one of the reactions below will occur?
- A) $\text{Zn}(s) + \text{MnI}_2(aq) \rightarrow \text{ZnI}_2(aq) + \text{Mn}(s)$
 - B) $\text{SnCl}_2(aq) + \text{Cu}(s) \rightarrow \text{Sn}(s) + \text{CuCl}_2(aq)$
 - C) $2\text{AgNO}_3(aq) + \text{Pb}(s) \rightarrow 2\text{Ag}(s) + \text{Pb}(\text{NO}_3)_2(aq)$
 - D) $3\text{Hg}(l) + 2\text{Cr}(\text{NO}_3)_3(aq) \rightarrow 3\text{Hg}(\text{NO}_3)_2(aq) + 2\text{Cr}(s)$
 - E) $3\text{FeBr}_2(aq) + 2\text{Au}(s) \rightarrow 3\text{Fe}(s) + 2\text{AuBr}_3(aq)$
- A11). A 0.200 M K_2SO_4 solution is produced by _____.
- A) dilution of 250.0 mL of 1.00 M K_2SO_4 to 1.00 L
 - B) dissolving 43.6 g of K_2SO_4 in water and diluting to a total volume of 250.0 mL
 - C) diluting 20.0 mL of 5.00 M K_2SO_4 solution to 500.0 mL
 - D) dissolving 20.2 g of K_2SO_4 in water and diluting to 250.0 mL, then diluting 25.0 mL of this solution to a total volume of 500.0 mL
 - E) dilution of 1.00 mL of 250 M K_2SO_4 to 1.00 L



A12). The internal energy for $\text{Mg}(s)$ and $\text{Cl}_2(g)$ is greater than that of $\text{MgCl}_2(s)$. Choose the energy diagram that represents the reaction $\text{MgCl}_2(s) \rightarrow \text{Mg}(s) + \text{Cl}_2(g)$.



A13). An electron cannot have the quantum numbers $n = \underline{\hspace{2cm}}$, $l = \underline{\hspace{2cm}}$, $m_l = \underline{\hspace{2cm}}$.

- A) 2, 0, 0
- B) 2, 1, -1
- C) 3, 1, -1
- D) 1, 1, 1
- E) 3, 2, 1

A14). The lowest orbital energy is reached when the number of electrons with the same spin is maximized. This statement describes _____.

- A) Pauli Exclusion Principle
- B) Planck's constant
- C) deBroglie hypothesis
- D) Heisenberg Uncertainty Principle
- E) Hund's rule

A15). A mole of red photons of wavelength 725 nm has _____ kJ of energy.

- A) 2.74×10^{-19}
- B) 4.56×10^{-46}
- C) 6.05×10^{-3}
- D) 165
- E) 227

A16). What is the de Broglie wavelength (m) of a 2.0-kg object moving at a speed of 50 m/s?

- A) 6.6×10^{-36} m
- B) 1.5×10^{35} m
- C) 5.3×10^{-33} m
- D) 2.6×10^{-35} m
- E) 3.8×10^{34} m

A17). Why does cesium react more violently with water than sodium?

- A) Sodium has greater metallic character than does cesium.
- B) The first ionization energy of cesium is less than that of sodium.
- C) The electron affinity of sodium is smaller than that of cesium.
- D) The effective nuclear charge for cesium is less than that of sodium.

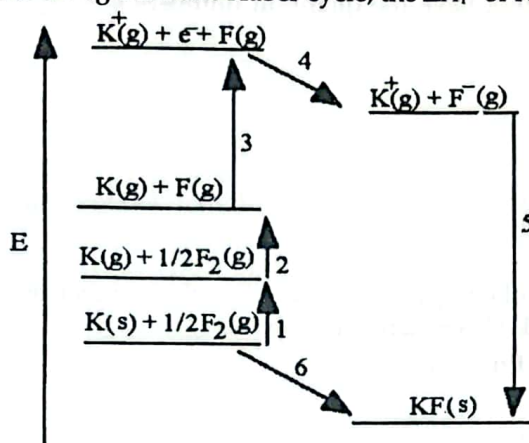


E) The atomic radius of cesium is smaller than that of sodium.

A18). Hypothetical elements X and Y form a molecule XY_2 , in which both Y atoms are bonded to atom X (and not to one another). The X—X distance in the elemental form of X is 2.04 Å, and the Y—Y distance in elemental Y is 1.68 Å. What would you predict for the X—Y distance in the XY_2 molecule?

- A) 0.84 Å
- B) 1.02 Å
- C) 1.86 Å
- D) 2.70 Å
- E) 3.72 Å

A19). The diagram below is the Born-Huber cycle for the formation of crystalline potassium fluoride. Using the Born-Haber cycle, the ΔH_f° of KF is equal to _____.



- A) $\Delta H_f^\circ[K(g)] + \Delta H_f^\circ[F(g)] + I_1(K) + E(F) + \Delta H_{\text{lattice}}$
- B) $\Delta H_f^\circ[K(g)] - \Delta H_f^\circ[F(g)] - I_1(K) - E(F) - \Delta H_{\text{lattice}}$
- C) $\Delta H_f^\circ[K(g)] - \Delta H_f^\circ[F(g)] + I_1(K) - E(F) + \Delta H_{\text{lattice}}$
- D) $\Delta H_f^\circ[K(g)] + \Delta H_f^\circ[F(g)] - I_1(K) - E(F) + \Delta H_{\text{lattice}}$
- E) $\Delta H_f^\circ[K(g)] + \Delta H_f^\circ[F(g)] + I_1(K) + E(F) - \Delta H_{\text{lattice}}$

A20). Use the VSEPR model to predict the electron-domain and molecular geometries for BrF_5 .

- | | |
|-------------------------|------------------|
| A) octahedral | square pyramidal |
| B) trigonal bipyramidal | linear |
| C) tetrahedral | bent |
| D) trigonal bipyramidal | T-shaped |
| E) octahedral | square planar |



Section B: Short Answer - Write legibly and show all work for all steps in the problem.
(24 Pts)

Number of questions: 6

Points per question: 4

B1). Give the chemical names of each of the following familiar compounds: (a) HNO_3 , (b) NaOH , (c) N_2O_4 , (d) Fe_2O_3 .

B2). Classify these dissolved substances as a strong electrolyte, weak electrolyte, or nonelectrolyte: $\text{Ca}(\text{OH})_2$, HNO_2 , $\text{C}_2\text{H}_5\text{OH}$ (ethanol), HCOOH (formic acid), HI , $\text{Ba}(\text{OH})_2$.

B3). The following electron configurations represent excited states: (a) $1s^2 2s^2 2p^4 3s^1$, (b) $[\text{Ar}] 4s^1 3d^{10} 4p^2 5p^1$, (c) $[\text{Kr}] 5s^2 4d^2 5p^1$, (d) $[\text{Ar}] 4s^1 3d^{10} 4p^4 5p^1$. Identify the element and indicate unpaired electrons in ground-state.

B4). Consider the first ionization energy of neon and the electron affinity of fluorine. (a) Write equations, including electron configurations, for each process. (b) Would you expect the magnitudes of these two quantities to be equal? If not, which one would you expect to be larger?

B5). Arrange the following substances in order of
(a) increasing electronegativity: Ga, P, As, S, O, F;
(b) decreasing polarity: $\text{Be}-\text{F}$, $\text{Mg}-\text{F}$, $\text{N}-\text{F}$, $\text{O}-\text{F}$;
(c) decreasing lattice energy: NaCl , MgO , CsI , ScN ;
(d) increasing atomic radius: Be, C, K, and Ca.

B6). In the formate ion, HCO_2^- , the carbon atom is the central atom with the other three atoms attached to it. (a) Draw the Lewis structures for the formate ion. (b) What hybridization is exhibited by the C atom? (c) How many electrons are in the π system of the ion?

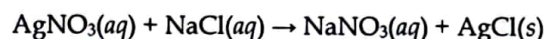


Section C: LongAnswer-- Write legibly and show all work for all steps in the problem. For mathematical relationships show both the equations used and substitution of values into the equation. Maintain correct units throughout your calculations and report your answers with the correct significant figures. (36 Pts)

Number of questions: 4 questions

Points per question: 9

C1). The precipitation reaction between $\text{AgNO}_3(aq)$ and $\text{NaCl}(aq)$ proceeds as follows at 25°C :



(a) Write the net ionic equation of this reaction. (b) Given the data in the table below, calculate ΔH° for the net ionic equation of this reaction. (c) What would you expect for the value of ΔH° of the overall molecular equation compared to that for the net ionic equation? Explain. (d) Use the results from (b) and (c) along with data the table below to determine the value of ΔH_f° for $\text{AgNO}_3(aq)$.

Compound	$\text{AgNO}_3(s)$	$\text{Ag}^+(aq)$	$\text{AgCl}(s)$	$\text{Cl}^-(aq)$	$\text{NaCl}(aq)$	$\text{NaNO}_3(aq)$
ΔH_f° (kJ/mol)	-124.4	105.90	-127.0	-167.2	-407.1	-446.2

C2). Titanium metal requires a photon with a minimum energy of $6.94 \times 10^{-19} \text{ J}$ to emit electrons. (a) What is the minimum frequency of light necessary to emit electrons from titanium via the photoelectric effect? (b) What is the wavelength of this light? (c) Is it possible to eject electrons from titanium metal using visible light, which wavelength ranges from 400 nm to 750 nm? (d) If titanium is irradiated with light of wavelength 233 nm, what is the maximum possible kinetic energy of the emitted electrons? (e) What is the maximum number of electrons that can be freed by a burst of light whose total energy is $2.00 \mu\text{J}$?

C3). (a) Draw the Lewis structure(s) for each of the following molecules and ions, and predict their molecular geometries: (i) CO_2 , (ii) NO_2^- , (iii) SnCl_3^- . (indicate all resonance structures if there are). (b) Indicate the hybridization of the central atom in (i) BCl_3 , (ii) AlCl_4^- , (iii) CS_2 .

C4). (a) What does the term diamagnetism mean? (b) Draw the molecular orbital electron configurations and indicate the bond order for: N_2^{2-} , O_2^{2-} , Be_2^{2+} , C_2^- . (c) Would you expect the ions in part (b) to be diamagnetic or paramagnetic?



Appendices

Fundamental Constants*	
Atomic mass unit	1 amu = $1.660538921 \times 10^{-27}$ kg 1 g = $6.02214165 \times 10^{23}$ amu
Avogadro's number†	$N_A = 6.02214129 \times 10^{23}$ /mol
Boltzman's Constant	$k = 1.3806488 \times 10^{-23}$ J/K
Electron charge	$e = 1.602176565 \times 10^{-19}$ C
Faraday's constant	$F = 9.64853365 \times 10^4$ C/mol
Gas constant	$R = 0.0820582$ L-atm/mol-K = 8.3144621 J/mol-K
Mass of electron	$m_e = 5.4857990946 \times 10^{-4}$ amu = $9.10938291 \times 10^{-31}$ kg
Mass of neutron	$m_n = 1.008664916$ amu = $1.674927351 \times 10^{-27}$ kg
Mass of proton	$m_p = 1.007276466$ amu = $1.672621777 \times 10^{-27}$ kg
Pi	$\pi = 3.1415927$
Planck's constant	$h = 6.62606957 \times 10^{-34}$ J-s
Speed of light in vacuum	$c = 2.99792458 \times 10^8$ m/s

activity series of metals in aqueous solution	
metal	oxidation Reaction
Lithium	$\text{Li}(s) \rightarrow \text{Li}^+(aq) + e^-$
Potassium	$\text{K}(s) \rightarrow \text{K}^+(aq) + e^-$
Barium	$\text{Ba}(s) \rightarrow \text{Ba}^{2+}(aq) + 2e^-$
Calcium	$\text{Ca}(s) \rightarrow \text{Ca}^{2+}(aq) + 2e^-$
Sodium	$\text{Na}(s) \rightarrow \text{Na}^+(aq) + e^-$
Magnesium	$\text{Mg}(s) \rightarrow \text{Mg}^{2+}(aq) + 2e^-$
Aluminum	$\text{Al}(s) \rightarrow \text{Al}^{3+}(aq) + 3e^-$
Manganese	$\text{Mn}(s) \rightarrow \text{Mn}^{2+}(aq) + 2e^-$
Zinc	$\text{Zn}(s) \rightarrow \text{Zn}^{2+}(aq) + 2e^-$
Chromium	$\text{Cr}(s) \rightarrow \text{Cr}^{3+}(aq) + 3e^-$
Iron	$\text{Fe}(s) \rightarrow \text{Fe}^{2+}(aq) + 2e^-$
Cobalt	$\text{Co}(s) \rightarrow \text{Co}^{2+}(aq) + 2e^-$
Nickel	$\text{Ni}(s) \rightarrow \text{Ni}^{2+}(aq) + 2e^-$
Tin	$\text{Sn}(s) \rightarrow \text{Sn}^{2+}(aq) + 2e^-$
Lead	$\text{Pb}(s) \rightarrow \text{Pb}^{2+}(aq) + 2e^-$
Hydrogen	$\text{H}_2(g) \rightarrow 2 \text{H}^+(aq) + 2e^-$
Copper	$\text{Cu}(s) \rightarrow \text{Cu}^{2+}(aq) + 2e^-$
Silver	$\text{Ag}(s) \rightarrow \text{Ag}^+(aq) + e^-$
Mercury	$\text{Hg}(l) \rightarrow \text{Hg}^{2+}(aq) + 2e^-$
Platinum	$\text{Pt}(s) \rightarrow \text{Pt}^{2+}(aq) + 2e^-$
Gold	$\text{Au}(s) \rightarrow \text{Au}^{3+}(aq) + 3e^-$



Hydrogen	Helium
1	2
H	He
1.008	4.0026
Lithium	Neon
3	10
Li	Ne
6.94	18.998
Sodium	Chlorine
11	17
Na	Cl
22.990	35.45
Magnesium	Argon
24.305	20.180
Mg	Ar
27.980	39.948
Calcium	Bromine
20	35
Ca	Br
39.098	79.904
Potassium	Krypton
19	36
K	Kr
40.078(4)	83.798(2)
Rubidium	Xenon
37	54
Rb	Xe
85.468	131.29
Cesium	Radium
55	86
Cs	Ra
132.91	226.0254
Francium	Ununennium
87	118
Fr	Uue
223.021(7)	289(1)
Beryllium	Fluorine
4	9
Be	F
9.0122	18.9984032(3)
Magnesium	Oxygen
12	8
Mg	O
24.304	15.999
Calcium	Sulfur
20	16
Ca	S
40.078	32.06
Potassium	Selenium
19	34
K	Se
39.098	78.96(3)
Rubidium	Tellurium
37	52
Rb	Te
85.468	127.60(3)
Cesium	Polonium
55	84
Cs	Po
132.91	209
Francium	Livermorium
87	116
Fr	Lv
223.021(7)	293(1)
Boron	Nitrogen
6	7
B	N
10.81	14.007
Aluminum	Phosphorus
13	15
Al	P
26.982	30.974
Gallium	Arsenic
31	33
Ga	As
69.723	74.922
Indium	Antimony
49	51
In	Sb
114.82	121.76
Thallium	Bismuth
81	83
Tl	Bi
204.38	208.98
Ununium	Ununpentium
113	115
Uut	Uup
284(18)	289(19)
Cadmium	Zinc
48	30
Cd	Zn
112.41	65.38(2)
Mercury	Copper
80	29
Hg	Cu
200.59	63.546(3)
Copernicium	Nickel
112	28
Cn	Ni
285(17)	58.693
Roentgenium	Palladium
111	46
Rg	Pd
280(16)	106.42
Darmstadtium	Platinum
110	78
Ds	Pt
281(16)	195.08
Meitnerium	Rhodium
109	45
Mt	Rh
278(15)	101.07(2)
Hassium	Osmium
108	76
Hs	Os
277(15)	190.23(2)
Bohrium	Iridium
107	77
Bh	Ir
270(1)	192.22
Seaborgium	Moscovium
106	115
Sg	Mc
271(13)	289(11)
Dubnium	Bohrium
105	107
Db	Bh
266(13)	270(1)
Rutherfordium	Seaborgium
104	106
Rf	Sg
261(12)	271(13)
Lr	Lu
262(11)	174.967(1)
Lawrencium	Rutherfordium
103	104
Lr	Rf
262(11)	261(12)
Unbinilium	Unbinilium
120	120
Ubn	Ubn
304(10)	304(10)

Lanthanum	57	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Yttrium
138.91	140.12	140.91	144.24	144.91	150.36(2)	151.96	157.25(3)	158.93	162.50	164.93	167.26	168.93	170.26	173.05	175.07	177.05
Actinium	89	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	101
227.03	232.04	232.04	238.03	237.05	244.06	244.06	247.07	251.08	252.08	257.10	259.10	261.10	265.11	267.11	269.11	271.10

***lanthanoids**

actinoids

